



Thèse

2025

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Relation entre morphologie cranio-faciale et caractéristiques anatomiques des muscles de la mastication : une revue systématique et méta-analyse

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How to cite

TOGNINALLI, David. Relation entre morphologie cranio-faciale et caractéristiques anatomiques des muscles de la mastication : une revue systématique et méta-analyse. Doctoral Thesis, 2025. doi: 10.13097/archive-ouverte/unige:182803

This publication URL: <https://archive-ouverte.unige.ch/unige:182803>

Publication DOI: [10.13097/archive-ouverte/unige:182803](https://doi.org/10.13097/archive-ouverte/unige:182803)

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**UNIVERSITÉ
DE GENÈVE**



**UNIVERSITÉ
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FACULTÉ DE MÉDECINE

Section de médecine dentaire
Département de réhabilitation oro-faciale
Division d'orthodontie

Thèse préparée sous la direction du Dre Alexandra Papadopoulou et du Pr Gregory Antonarakis

***Relation entre morphologie cranio-faciale et
caractéristiques anatomiques des muscles de la
mastication : une revue systématique et méta-analyse***

Thèse
présentée à la Faculté de Médecine
de l'Université de Genève
pour obtenir le grade de Docteur en médecine
par

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de
Genève (GE)

Thèse n° 818

Genève

2025

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Résumé

Objectif : Identifier les différences dans les caractéristiques anatomiques des muscles masticatoires selon les différentes morphologies cranio-faciales, sur le plan vertical (divergence) ainsi que sur le plan sagittal (classe squelettique).

Sujets et Méthodes : Une recherche de la littérature a été menée jusqu'à décembre 2023 sur plusieurs bases de données électroniques telles que PubMed, Embase, Web of Science, SciELO, Scopus, ProQuest, VHL, et Cochrane Library. En parallèle, une recherche manuelle a été réalisée. L'objectif de cette recherche était de recenser toutes les études publiées examinant la relation entre les caractéristiques des muscles masticatoires et les différentes morphologies cranio-faciales. Après l'extraction des données et l'évaluation du risque de biais pour chaque étude, une analyse qualitative ainsi qu'une analyse quantitative ont été menées pour explorer le lien entre la morphologie faciale squelettique et les caractéristiques musculaires (épaisseur, section transversale et volume).

Résultats : Trente-quatre études (37 publications) ont été incluses, représentant un total de 2'047 participants analysés avec différentes méthodes d'imagerie (ultrasons, Cone-Beam CT, IRM et CT scans). Les données de 16 études (804 patients) ont été utilisées pour la méta-analyse et les résultats montrent que l'épaisseur du muscle masséter est significativement plus importante de 1.14 mm (95% CI 0.74 à 1.53 mm) chez les patients hypodivergents par rapport aux sujets normodivergents. Inversement les patients hyperdivergents ont une épaisseur du muscle masséter significativement moins importante de - 1.14 mm (95% CI -1.56 à -0.73 mm) comparé aux patients normodivergents. Les mêmes observations peuvent être faites pour les mesures du muscle masséter prises lors de la contraction musculaire maximale, ainsi que pour sa section transversale et son volume au repos.

Conclusions : Des variations significatives du muscle masséter ont été observées en fonction des divergences cranio-faciales des patients, bien que les études incluses dans la revue systématique et la méta-analyse présentent un risque élevé de biais.

Introduction

L'apparence faciale revêt une importance croissante dans la société contemporaine. Cela se reflète dans l'augmentation du nombre de patients ayant recours à des interventions de chirurgie esthétique ou à des injections. Cette tendance est particulièrement marquée dans le domaine de la dentisterie, où un nombre croissant de patients recherche des traitements dentaires esthétiques tels que des facettes ou des traitements orthodontiques. L'industrie de l'orthodontie progresse rapidement avec sa transformation digitale et est de plus en plus agressive envers nos patients avec de la publicité parfois mensongère [1], donnant naissance à des nouvelles formes de traitements tels que le "direct-to-consumer orthodontics" [2,3]. Cependant, l'orthodontie ne se limite pas à alignement des dents, mais vise également à guider et coordonner la croissance cranio-faciale dès le plus jeune âge, souvent à l'aide d'appareils orthopédiques. Le diagnostic et le traitement des problèmes orthodontiques font partie à part entière d'une prise en charge médicale et recouvrent une multitude de différentes malocclusions, allant parfois d'un simple encombrement dentaire à des dysmorphoses cranio-faciales importantes nécessitant une prise en charge combinée avec un traitement de chirurgie maxillo-faciale [4,5]. Pour cette raison, il est de notre devoir en tant que médecins dentistes spécialistes d'être aussi compétents que possible dans notre domaine afin de fournir le meilleur traitement à nos patients. Dans cette optique, il est crucial de comprendre les mécanismes de la croissance cranio-faciale qui mènent à la morphologie de chacun, ainsi que les facteurs qui l'influencent. En effet, le développement et la croissance des structures faciales, tels que les bases osseuses et les tissus mous, notamment les muscles, les nerfs, et les voies aéro-digestives supérieures, revêtent un intérêt primordial pour l'orthodontiste.

Croissance cranio-faciale :

La croissance cranio-faciale commence in utero au cours de l'embryogenèse avec la migration et la différenciation des cellules de la crête neurale. L'organisation des tissus et les modèles de croissance sont fortement dictés par l'expression de gènes qui s'activent et s'inhibent mutuellement [6]. La croissance du massif facial suit les patterns de croissance somatique : durant les premières années de vie du nouveau-né, cette croissance est relativement marquée et diminue vers l'âge de 6 ans, pour reprendre lors de l'adolescence jusqu'à l'âge adulte (Figure 1) [7]. En revanche, la voûte crânienne suit un pattern de croissance différent dicté par la croissance du tissu cérébral qu'elle contient : elle atteint très rapidement sa taille adulte, généralement avant l'âge de 10 ans (Figure 2).

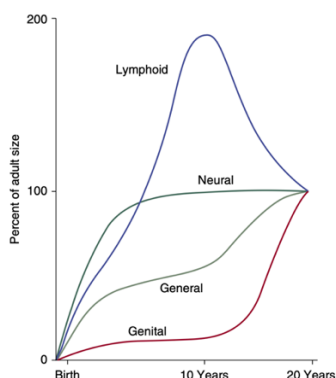


Figure 1 Courbe de Scammon illustrant le fait que différents systèmes du corps ont des taux de développement différents et arrivent à maturité à des âges différents (adapté de Scammon RD. *The measurement of the body in childhood*. In : Harris JA, ed. *The Measurement of Man*. Minneapolis : University of Minnesota Press ; 1930)

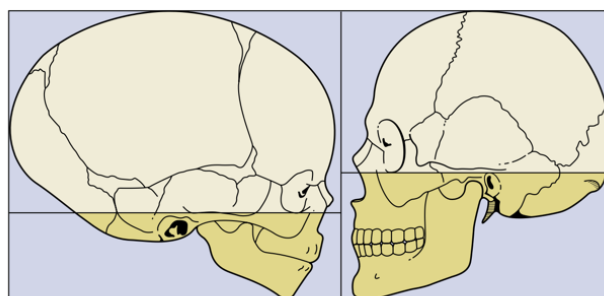


Figure 2 Disproportions de la tête et du visage chez le nourrisson et l'adulte. Le neurocrâne, qui abrite le cerveau et les yeux est précoce dans son développement et sa croissance est donc proportionnellement plus grande que celle du visage pendant la petite enfance. Pendant l'adolescence, la croissance du massif facial est plus importante (adapté de Lowery GH. *Growth and Development of Children*. 6e éd. Chicago : Year Book Medical Publishers ; 1973.)

Le complexe nasomaxillaire (tiers moyen du visage) est composé de différentes structures osseuses tels que l'os maxillaire, l'os nasal, l'os zygomatique, l'os lacrymal, l'os palatin et l'os vomer. Sa croissance se fait par ossification intra-membraneuse au niveau des sutures frontozygomatiques, frontomaxillaires, frontonasales, zygomatocomaxillaires, temporozygomatique, pterygopalatine, et nasomaxillaires entraînant un déplacement de tout le complexe vers le bas et l'avant (Figure 3-4).

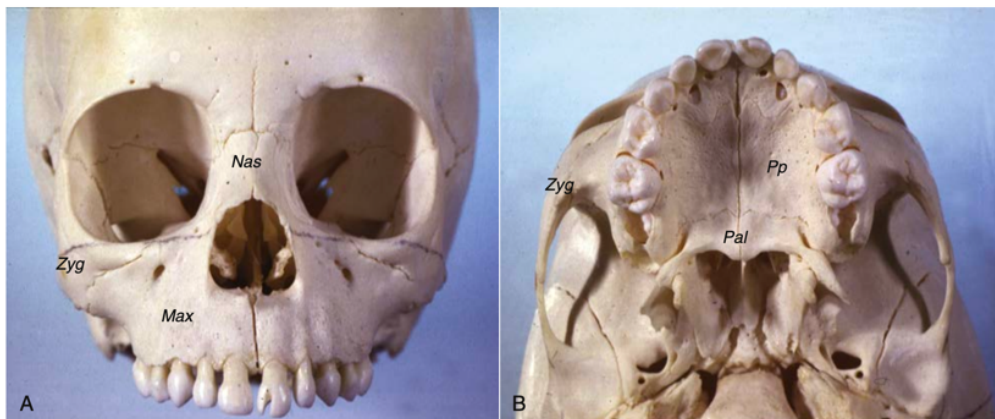


Figure 3 (A) Vues frontales et (B) basales d'un jeune humain indiquant les os comprenant le milieu du visage. Max, maxillaire ; Nas, os nasaux ; Zyg, os zygomatiques ; Pal, os palatins ; Pp, processus palatins des os maxillaires.

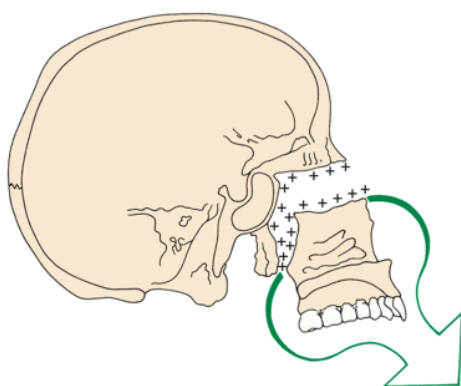


Figure 4 Au fur et à mesure que la croissance des tissus mous environnants déplace le maxillaire vers le bas et vers l'avant, ouvrant un espace au niveau de ses attaches suturales supérieures et postérieures, de l'os nouveau est ajouté de part et d'autre des sutures. (adapté de Enlow DH, Hans MG. *Essentials of Facial Growth*. Philadelphie : WB Saunders ; 1996).

La mandibule (tiers inférieur du visage) grandit quant à elle au niveau du cartilage de ses condyles (croissance endochondrale). La croissance des condyles est dirigée vers le haut et vers l'arrière, ce qui provoque un déplacement de tout le corps mandibulaire vers le bas et l'avant, comme pour le maxillaire (Figure 5).

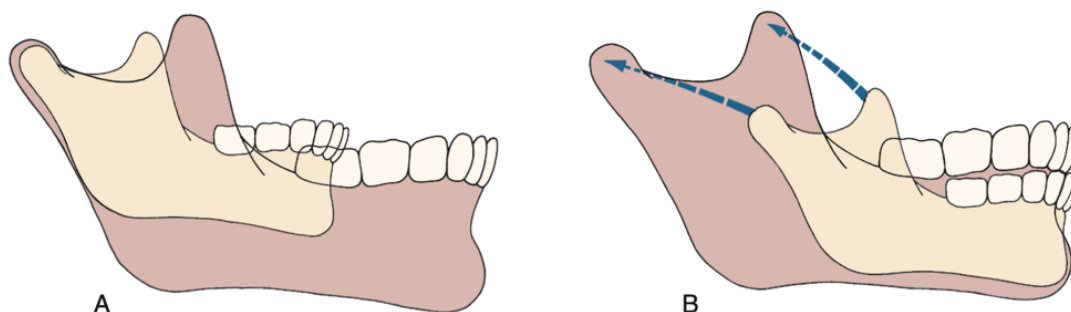


Figure 5 (A) Croissance de la mandibule, vue sous l'angle d'une base crânienne stable. Le menton se déplace vers le bas et vers l'avant. (B) Croissance de la mandibule, vue sous l'angle des études de coloration vitale, qui révèlent des changements minimes dans la zone du corps et du menton, alors qu'il y a une croissance et un modelage exceptionnels du ramus, le déplaçant vers l'arrière. Le concept correct est que la mandibule est translatée vers le bas et vers l'avant et qu'elle croît vers le haut et vers l'arrière en réponse à cette translation, en maintenant son contact avec le crâne.

Le remodelage des surfaces osseuses est aussi responsable de la morphologie faciale, au maxillaire on retrouve, lors de la croissance, une zone de résorption au niveau des zones paranasales en surface ainsi qu'au niveau du plancher de la cavité nasale et une zone d'apposition au niveau du palais (Figure 6) [8]. Pour la mandibule la résorption se fait au niveau antérieur de la branche montante et l'apposition se fait sur la zone postérieure (Figure 7) [9].

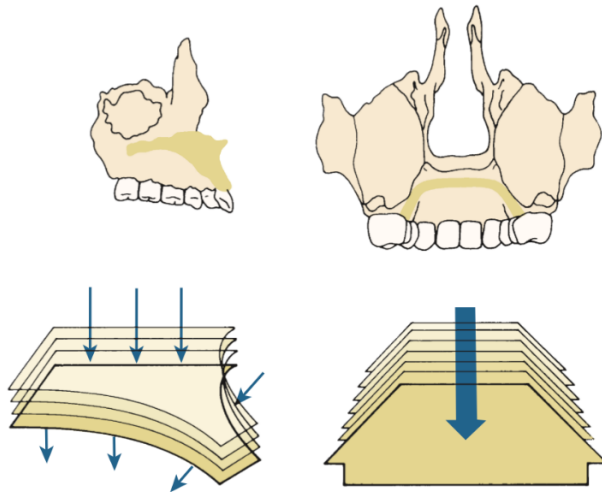


Figure 6 La modélisation de la voûte palatine (qui est aussi le plancher du nez) la déplace dans la même direction que la translation ; de l'os est retiré du plancher du nez et ajouté au toit de la bouche. Sur la face antérieure, cependant, de l'os est enlevé, ce qui annule partiellement la translation vers l'avant. Lorsque la voûte se déplace vers le bas, le même processus de remodelage osseux l'élargit également. (adapté de Enlow DH, Hans MB. *Essentials of Facial Growth*. Philadelphie : WB Saunders ; 1996).

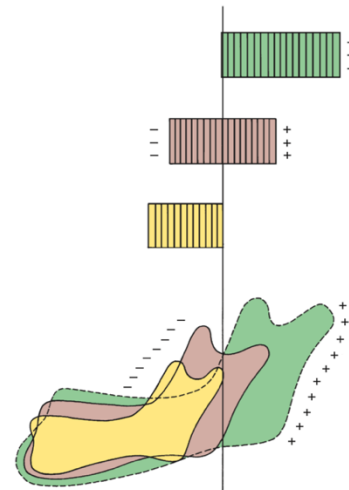


Figure 7 Au fur et à mesure que la mandibule s'allonge, le ramus est largement modelé, à tel point que l'os qui se trouve à l'extrémité du processus condylien à un âge précoce peut se retrouver à la surface antérieure du ramus quelques années plus tard. (adapté de Enlow DH, Hans MB. *Essentials of Facial Growth*. Philadelphie : WB Saunders ; 1996).

La croissance cranio-faciale est donc intrinsèquement guidée par l'expression des gènes, mais elle peut également être modifiée par des facteurs environnementaux tels que les muscles, les fonctions, les traumatismes et les mauvaises habitudes.

Muscles de la mastication et croissance cranio-faciale :

Les muscles de la mastication peuvent être classés d'un point de vue clinique en deux grands groupes : les muscles élévateurs et les muscles dépresseurs. Le premier groupe comprend le muscle masséter, le muscle temporal (aussi impliqué dans la rétro-pulsion de la mandibule) ainsi que le muscle ptérygoïde médial et le chef supérieur du muscle ptérygoïde latéral. Le

deuxième groupe est lui composé du muscle digastrique antérieur, mylohyoïdien, et le chef inférieur du muscle ptérygoïde latéral [10].

Cependant, la littérature se concentre principalement sur les muscles masticateurs élévateurs de la mandibule, en particulier le muscle masséter, et leur relation avec les différentes morphologies cranio-faciales. L'échographie (US) est la méthode la plus couramment utilisée pour évaluer l'épaisseur de ces muscles, en raison de ses nombreux avantages tels que l'absence de rayonnement ionisant, son coût relativement bas et sa rapidité [11–15].

Il est désormais scientifiquement prouvé que les muscles et les contraintes qu'ils exercent sur les structures osseuses ont une influence importante sur leur remodelage et leur morphologie [16–19]. La fonction des muscles masticateurs est directement liée à l'accroissement du stress sur les mâchoires et à la formation osseuse, ce qui peut influencer des zones, tels que l'angle gonial [19]. Cet angle joue un rôle crucial en servant de point d'attache pour des muscles majeurs tels que le masséter et le ptérygoïdien médial. Ainsi, les variations de taille et d'activité de ces muscles pourraient impacter la configuration de l'angle gonial et, par extension, la morphologie dento-faciale globale.

La loi de Wolff énonce que la structure osseuse est influencée par l'épaisseur des muscles, suggérant ainsi un lien intrinsèque entre la fonction musculaire et la forme du squelette interne [20]. En appliquant ce principe, plusieurs études évoquent une corrélation négative entre l'épaisseur du muscle masséter et la morphologie faciale verticale ainsi que la longueur du visage [21–28]. Les mesures céphalométriques telles que l'angle du plan mandibulaire et l'angle gonial montrent une corrélation négative avec l'épaisseur du muscle masséter, tandis que la hauteur de la branche montante mandibulaire et la hauteur faciale postérieure présentent une corrélation positive avec cette épaisseur musculaire.

Les variations dans l'orientation et l'insertion des fibres musculaires sont souvent associées à des morphologies dento-faciales distinctes. Chez les individus présentant un motif squelettique hypodivergent (visage court), la chaîne verticale postérieure des muscles est orientée de manière plus verticale et insérée de manière plus antérieure que chez ceux présentant un motif squelettique hyperdivergent (visage long) [29]. Selon Takada et al. [30], les muscles masséters d'orientation verticale (attachés en avant sur la mandibule) sont associés à une insertion musculaire masséter inférieure par rapport au plan occlusal, ainsi qu'à une face postérieure longue, un plan mandibulaire plat et un angle gonial aigu, caractéristiques observées chez les enfants hypodivergents. En revanche, Proctor et DeVincenzo [31] ont observé des muscles masséters plus horizontal par rapport à la base crânienne, au plan horizontal de Francfort et au plan palatin chez les individus hyperdivergents.

Lorsqu'on évalue les forces masticatoires chez des adultes, ceux ayant des muscles masticatoires moins puissants, et donc plus fins, présentent cliniquement un visage dolichocéphale (long). En revanche, les sujets dotés de muscles masticateurs plus forts ont tendance à avoir un aspect facial plutôt brachycéphale (court) [19,32]. Une différence de force musculaire masticatoire entre les individus normodivergents et hyperdivergents n'est pas évidente chez les enfants âgés de 6 à 11 ans, ce qui suggère une éventuelle incapacité des muscles élévateurs mandibulaires à augmenter en force chez les individus hyperdivergents en croissance [33].

D'autres études ont examiné la relation entre l'épaisseur musculaire et les dimensions cranio-faciales : Van Spronsen et al. [34] met en évidence la corrélation significative entre la surface transversale du muscle temporal antérieur et la largeur de la tête et du visage. Kitai et al. [35] suggèrent que les variables musculaires sont significativement corrélées à la largeur de l'arc bizygomatique et de la fosse temporale mais non corrélées à la largeur du crâne. Chan et al.

[36] ont montré que les sujets en croissance dotés de muscles masticateurs plus épais sont plus susceptibles d'avoir une plus grande largeur d'arc bizygomatique.

Conditions affectants la croissance cranio-faciale :

Dans certaines situations, la morphologie faciale peut être altérée en raison de maladies congénitales ou acquises, perturbant le processus de croissance. De plus, des dysfonctionnements des fonctions masticatoires ou respiratoires peuvent également contribuer à une altération de la morphologie faciale.

Myopathie de Duchenne :

Il s'agit d'une maladie génétique qui affecte principalement les garçons, avec une prévalence d'environ 7 cas sur 100'000 garçons [37]. Elle se manifeste généralement vers l'âge de 4 ans et se caractérise par une dégénérescence musculaire progressive. Les disparités dans la capacité musculaire se traduisent par des variations dans l'épaisseur des muscles. Katsaros et al. [38] ont montré que les patients atteints de maladies neuromusculaires dégénératives présentent des muscles moins développés et moins puissants, ce qui influe directement sur la morphologie cranio-faciale, entraînant une croissance verticale (hyperdivergent) plutôt qu'horizontale (hypodivergent).

Arthrite juvénile idiopathique (AJI) :

L'arthrite juvénile idiopathique est la maladie rhumatismale chronique la plus commune chez les enfants, touchant en Europe 60 sujets pour 100'000 habitants [39]. Ce terme désigne un ensemble de maladies inflammatoires chroniques qui se manifestent toutes avant l'âge de 16 ans, avec une durée supérieure à six semaines. Elles se caractérisent par des douleurs articulaires persistantes, un gonflement, une sensation de chaleur et raideur, généralement exacerbés le matin et après une position assise prolongée [40]. L'inflammation des articulations mène à une altération de la croissance des membres mais aussi de la structure

cranio-faciale. En effet, selon l'étude de Twilt et al. [41], les patients atteints d'AJI présentent un risque de 67 % de développer une rétrognathie et de 52 % d'avoir une rotation postérieure de la mandibule. Lorsque les articulations temporo-mandibulaires sont affectées par l'inflammation chronique, ces chiffres augmentent à 82 % pour la rétromandibulie et à 58 % pour la rotation postérieure. Les analyses céphalométriques révèlent ainsi une augmentation de la hauteur du tiers inférieur du visage, accompagnée d'une croissance verticale plus prononcée (patients hyperdivergents). On observe une rotation dans le sens horaire de la mandibule, associée à une tendance à l'ouverture de l'occlusion dentaire (openbite) [42].

Végétations adénoïdes :

La fonction respiratoire joue un rôle prédominant dans la croissance cranio-faciale. Certains patients présentent une respiration buccale qui est due le plus souvent due à une obstruction nasale. La cause principale de cette ventilation orale est l'hypertrophie des végétations situées dans le rhinopharynx.

Comme l'on peut observer sur la courbe de Scammon (Figure 1) [7], ce tissu lymphoïde se développe de manière importante au cours des premières années de vie, atteignant son pic avant l'âge de 10 ans, pour ensuite régresser naturellement pendant l'adolescence. Si les végétations sont trop volumineuses, ou ne régressent que partiellement, l'on parle de végétations adénoïdes.

La persistance des végétations adénoïdes favorise une ouverture permanente de la bouche et une position plus basse de la langue dans la cavité orale. Un déséquilibre des forces exercées par les tissus mous sur les structures osseuses du complexe cranio-facial est observé [43], ce qui amène au *faciès adénoïdien*.

Les patients présentent une incompetence labiale, un maxillaire supérieur étroit, des incisives inférieure rétroclinées, une augmentation de la hauteur faciale antérieure, un angle

mandibulaire important ainsi qu'une rétromandibulie [44]. Ces mêmes constatations ont déjà été faites sur des expérimentations animales (primates) en obstruant leur respiration nasale dès les années 1980 par le Dr. Harvold [45].

Apnées obstructives du sommeil :

Chez tous les individus, le tonus musculaire des voies aériennes diminue pendant le sommeil [46], ce qui n'a généralement peu d'impact sur des sujets sains. En revanche, si les patients présentent des facteurs de risque des apnées obstructives du sommeil peuvent survenir. Les principaux facteurs de risque pour l'enfant sont les végétations adénoïdes, l'obésité et certaines morphologies cranio-faciales ou syndromes génétiques. La prévalence des apnées obstructives du sommeil pédiatriques est de 1 à 4 % [47].

Zucconi et al. [48] ont comparé les caractéristiques céphalométriques de 26 enfants âgés de 5 ans souffrant d'apnées obstructives du sommeil avec un groupe contrôle. Les résultats montrent qu'ils ont une morphologie squelettique très similaires à celles des enfants souffrant de végétations adénoïdes : ils présentent un angle mandibulaire ouvert, une hauteur faciale antérieure augmentée et une hauteur faciale postérieure diminuée (croissance de la branche montante mandibulaire moins importante).

Accidents et croissance cranio-faciale :

Comme évoqué dans les paragraphes précédents, les contraintes mécaniques, la fonction respiratoire, les conditions innées ou acquises, ainsi que les tissus mous jouent tous un rôle prédominant sur la croissance cranio-faciale. Cependant, il est important à ce stade de soulever la différence entre *centre* et *lieux* de croissance : tous les centres sont des lieux de croissance, mais l'inverse ne vaut pas.

Trois théories à ce jour ont tenté d'expliquer d'où vient la régulation génétique de la croissance du massif facial (centres de croissance) : (1) la croissance est régulée par les sutures, au niveau

des structures osseuses, (2) la croissance est régulée par le cartilage et (3) la croissance est régulée par les tissus mous. Baume [49] a pour la première fois introduit ces termes et dans sa publication de 1961 où il étudie des coupes histologiques de têtes de fœtus il conclut que les synchondroses sont des centres de croissance tout comme le sont les zones épiphysaires des os longs. La deuxième théorie est donc confirmée par Koski et al. [50] en 1970 qui parviennent à transplanter sur des rats du cartilage prélevé des synchondroses de la base du crâne et observent que l'os adjacent au cartilage continue sa croissance dans le site transplanté, témoignant de son pouvoir inné de croissance. Le même genre d'expérimentations ont été menées pour les sutures mais sans résultats satisfaisants, écartant donc la première théorie [51].

Suite à ces considérations le potentiel de croissance du massif facial peut facilement être altéré lors d'épisodes de traumatismes importants. Un endommagement du cartilage du septum nasal chez un jeune enfant peut entraîner une hypoplasie maxillaire à l'âge adulte. Une fracture du condyle, en revanche, lorsqu'elle est correctement prise en charge n'empêchera pas une bonne croissance mandibulaire dans 75% à 80% des cas [51]. Inversement, si le processus de guérison est compromis avec une immobilisation de l'articulation ou un tissu cicatriciel trop important, des séquelles telles qu'une ankylose de l'articulation temporo-mandibulaire ou un défaut de croissance mandibulaire peuvent survenir. De même, une altération de la croissance des structures osseuses peut être observée lors de tissus cicatriciels rigides des tissus mous, comme par exemple suite à une brûlure profonde et étendue qui redirige la croissance mandibulaire vers le bas entraînant une augmentation de la hauteur faciale inférieure [52,53].

Objectifs

Suite à cette introduction, il est évident que la morphologie faciale est influencée à la fois par des facteurs génétiques et environnementaux. Le domaine d'étude est vaste et passionnant. Bien que de nombreuses études aient été publiées sur le rôle des muscles de la mastication, moins d'attention a été portée à d'autres domaines.

L'objectif principal de cette revue systématique et méta-analyse était d'examiner de manière exhaustive les nombreuses preuves scientifiques disponibles à ce jour concernant la relation entre les caractéristiques anatomiques macroscopiques des muscles masticatoires et les différentes morphologies cranio-faciales (tant sur le plan sagittal que vertical), et de les synthétiser.

Les caractéristiques musculaires étudiées incluent l'épaisseur, la largeur, la surface transversale, le volume ainsi que l'orientation des muscles.

Les méthodes utilisées pour évaluer ces caractéristiques anatomiques incluent des imageries 2D ou 3D, telles que l'échographie (US), l'imagerie par résonance magnétique (IRM), la tomographie par faisceau conique (CBCT) et la tomodensitométrie conventionnelle (CT).

REVIEW

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Relationship between craniofacial skeletal patterns and anatomic characteristics of masticatory muscles: a systematic review and meta-analysis

David Togninalli¹, Gregory S. Antonarakis¹ and Alexandra K. Papadopoulou^{1,2*} **Abstract****Background** The anatomic characteristics of the masticatory muscles differ across craniofacial skeletal patterns.**Objective** To identify differences in the anatomic characteristics of masticatory muscles across different sagittal and vertical craniofacial skeletal patterns.**Eligibility criteria** Studies measuring the thickness, width, cross-sectional area (CSA), volume and orientation of masticatory muscles in healthy patients of different sagittal (Class I, Class II, and Class III) and/or vertical (normodivergent, hypodivergent, and hyperdivergent) patterns.**Information sources** Unrestricted literature searches in 8 electronic databases/registers until December 2023.**Risk of bias and synthesis of results** Study selection, data extraction, and risk of bias assessment with a customised tool were performed independently in duplicate. Random-effects meta-analysis and assessment of the certainty of clinical recommendations with the GRADE approach were conducted.**Results** 34 studies (37 publications) were selected with a total of 2047 participants and data from 16 studies were pulled in the meta-analysis. Masseter muscle thickness in relaxation was significantly greater by 1.14 mm (95% CI 0.74–1.53 mm) in hypodivergent compared to normodivergent patients while it was significantly decreased in hyperdivergent patients by –1.14 mm (95% CI –1.56 to –0.73 mm) and –2.28 mm (95% CI –2.71 to –1.85 mm) compared to normodivergent and hypodivergent patients respectively. Similar significant differences were seen between these groups in masseter muscle thickness during contraction as well as masseter muscle CSA and volume. Meta-analyses could not be performed for sagittal categorizations due to insufficient number of studies.**Conclusions** Considerable differences in masseter muscle thickness, CSA and volume were found across vertical skeletal configurations being significantly reduced in hyperdivergent patients; however, results should be interpreted with caution due to the high risk of bias of the included studies. These variations in the anatomic characteristics of masticatory muscles among different craniofacial patterns could be part of the orthodontic diagnosis and treatment planning process.**Registration:** PROSPERO [CRD42022371187](https://doi.org/10.1186/CRD42022371187).**Keywords** Masticatory muscles, Masseter, Craniofacial pattern, Skeletal pattern, Systematic review, Meta-analysis

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Introduction

Rationale

Craniofacial growth begins in-utero during embryogenesis with the migration and differentiation of neural crest cells. Tissue organization and growth patterns are highly dictated by the expression of genes involved in these processes. However, environmental factors can potentially alter this pre- or post-natal growth [1]. It has been shown that muscles and the stresses they exert on the bone can have an important influence on its remodelling and morphology [2–4]. Several studies have investigated the relationship between muscle thickness and craniofacial dimensions. Van Spronsen et al. [5] highlighted a significant correlation between the cross-sectional area of the anterior temporal muscle and facial width. Kitai et al. [6] suggested that the muscular variables are significantly correlated with the bizygomatic arch width and temporal fossa but are not correlated with the cranial width. Chan et al. [7] found that growing subjects with thicker masticatory muscles are more likely to have a greater bizygomatic arch width.

Differences in muscle fibre orientation and insertion are also related to different dentofacial morphologies with the masticatory muscles of individuals with a skeletal hypodivergent pattern being more uprightly oriented and anteriorly inserted compared to those with skeletal hyperdivergent patterns [8]. Takada et al. [9] suggest an association between vertically oriented masseter muscles that are anteriorly attached on the mandible and a long posterior face height accompanied with a flat mandibular plane and an acute gonial angle in hypodivergent children. On the other hand, Proctor and De Vincenzo report more horizontally oriented masseter muscles relative to the cranial base, Frankfort horizontal and the palatal plane in hyperdivergent individuals [10].

When masticatory forces were evaluated in adult individuals, patients with less powerful and thinner masticatory muscles were dolichocephalic. Conversely, subjects with strong masticatory muscles have a rather brachycephalic facial pattern [11, 12]. This difference in masticatory muscle strength between normodivergent and hyperdivergent individuals is not evident in children aged 6–11 years, indicating a possible inability of the mandibular elevator muscles to gain strength in the hyperdivergent group with growth [13]. Additionally, these differences in muscular force capacity are reflected in muscle thickness as patients affected by degenerative neuromuscular diseases have thinner muscles of lower strength that directly affect craniofacial morphology [14].

The function of the masticatory muscles is linked to increased stress on the jaws and the formation of bone, potentially influencing areas such as the gonial angle [11]. This part of the mandible serves as a point of attachment

for the masseter and medial pterygoid muscles, thus variations in their size and activity could impact mandibular shape and, consequently, overall dentofacial morphology. According to Wolff's law (1870), bone structure is influenced by muscle thickness, implying a connection between muscle function and the internal skeletal structure and form [15, 16]. An inverse correlation has been reported between masseter muscle thickness and anterior face height, mandibular plane angle and the gonial angle, while masseter muscle thickness was positively correlated with mandibular ramus height and posterior facial height [17–27].

Diverse methods have been used for the assessments of masticatory muscle attributes. Muscle biopsies have been employed to analyse characteristics such as muscle fibre type, composition and thickness, but due to its invasive nature is not used routinely [28]. Alternatively, muscular functional activity and force generation can be assessed with electromyography and bite force measurements respectively while muscular dimensions are measured using computer tomography, magnetic resonance imaging and ultrasonography. From the imaging techniques, ultrasonography (US) is the most widespread method for the analysis of the thickness of masticatory muscles for its advantages such as the absence of ionizing radiation, convenience, and rapidity [27, 29–32].

Objectives

The aim of this systematic review was to assess in a systematic manner the available evidence regarding the potential relationships between craniofacial patterns (sagittal and vertical) and masticatory muscle macroscopic anatomic characteristics, such as thickness (depth), width, cross-sectional area, volume and angle orientation assessed with any 2D or 3D imaging method.

Methods

Protocol registration, research question and eligibility criteria

The reporting of the present systematic review and meta-analysis is based on the PRISMA guidelines [33].

The study protocol was registered with the international prospective register of systematic reviews (PROSPERO CRD42022371187).

The research question was whether any differences exist between the size and/or orientation characteristics of the muscles of mastication in healthy humans and different craniofacial patterns (sagittal or vertical). The eligibility criteria were based on the PECO framework (population, exposure, comparator, outcomes) and are described in detail in Supplementary Table 1.

Information sources and search strategy

The databases, registers, websites, organizations, and other sources searched to identify potentially eligible studies as well as all search strategies are described in Supplementary Table 2. The last electronic literature search was performed on December 2023 and no limitations regarding publication year, language, status, or type were imposed (apart from filters for studies on humans, where they existed). Additionally, the reference lists of all included studies and all relevant systematic reviews were checked for additional studies. The literature search was carried out by two reviewers independently (DT and AKP).

Study selection

The selection of studies was performed by two independent reviewers (DT, AKP) based on screening titles and abstracts. The full-text versions of the pre-selected articles were accessed to assess their eligibility. Any disagreements were resolved through discussion while in the absence of consensus, a third reviewer (GSA) was consulted until consensus was reached. All relevant citations were imported to a reference manager software (End-Note® 20, Thomson Reuters, Philadelphia, PA) for deduplication. Researchers were not blinded to the authors of included studies.

Data collection and data items

The data collection procedure was carried out by two independent reviewers (DT and AKP) using pre-designed and pre-piloted forms (Supplementary Table 3) and extracted data were imported in digital spreadsheets. Discrepancies were resolved in the same way as above by consulting another author (GSA). Researchers were not blinded to the authors of included studies.

Study risk of bias assessment risk of bias within individual studies

A customized risk of bias tool was used for assessment of internal validity / reporting quality of each individual study independently and in duplicate by two investigators (DT and AKP). This tool was tailored to the scope of this review's eligible studies and based on items from The Joanna Briggs Institute's critical appraisal checklists for cross-sectional studies and the Appraisal tool for Cross-Sectional Studies (AXIS) [34, 35]. The specific domains / questions of the customized tool can be seen in Supplementary Table 4.

Summary measures, data synthesis and certainty assessment

The main objective of our systematic study and meta-analysis was to investigate the differences, if these exist, between the muscles of mastication and the sagittal or vertical craniofacial patterns in healthy humans. The Mean Difference (MD) with its 95% Confidence Interval (CI) was used to estimate anatomic characteristics in masticatory muscle differences among Class I, II and III or among normodivergent, hypodivergent and hyperdivergent patient groups.

As anatomic characteristics in masticatory muscles were expected to vary according to patient-related (chronological age, developmental growth stage, sex) or measurement-related factors (radiographical technical characteristics, method error, cut-off values used for categorization), a random-effects model was a priori deemed (using clinical / methodological justification [36]) most appropriate to incorporate this variability and estimate the average distribution of effects across studies.

The heterogeneity of the studies was determined using I² statistics as well as Tau² statistics [37, 38]. Pooled mean differences between groups (normo-, hypo- and hyperdivergent) were obtained with multivariate mixed-effects linear models for meta-analyses. Leave-one-out sensitivity analyses were conducted to check the robustness of the findings. The trim and fill approach was used to correct the pooled mean differences for a potential publication bias. Due to the low number of studies, the latest was used only on the most frequently reported outcome (masseter thickness). All statistical tests were two-sided with a significance level of 0.05. Statistical analyses were carried out with the package Metafor v3.8-1 for R v4.0.2 (R Core Team (2020). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL <https://www.R-project.org/> (<https://www.r-project.org/>)). The certainty of evidence (confidence in effect estimates) for both the primary and secondary outcomes as per the PECO table (Supplementary Table 1) was assessed using the GRADE approach [39].

Results

Study selection

A total of 3868 studies were retrieved from the databases, which after deduplication, selection according to title, abstract and full text were eliminated to 37 studies included in the present review. Studies were excluded in full text if they assessed outcomes different to the

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only

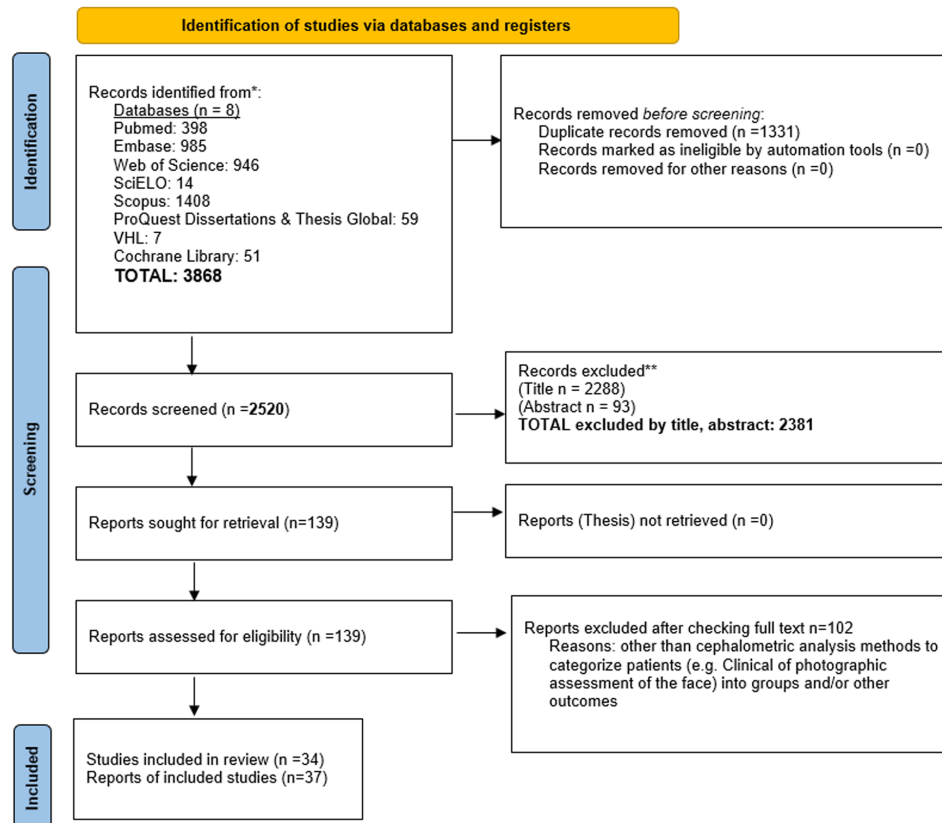


Fig. 1 PRISMA flowchart diagram

ones of interest or did not include at least two sagittal or vertical groups for comparison. The flow diagram of the studies retrieved from the databases is described in detail in the PRISMA diagram in Fig. 1, Supplementary Table 5. For the quantitative analysis, only 6 studies were included for the cross-sectional area (CSA) of the masseter, 7 for the volume of the masseter, 10 for the thickness of the masseter under relaxation, and 8 for the thickness of the masseter under contraction (during biting). The quantitative analysis focuses solely on patient divergence (hypodivergent vs. normodivergent vs. hyperdivergent), while for the sagittal relationship (Class I vs. Class II vs. Class III), an insufficient number

of studies (or subjects) have been published, preventing a meta-analysis from being conducted.

Study characteristics

Thirty-four (34) studies (37 publications) were included in the present review (28 prospective, 6 retrospective and 1 was of unclear study design). Muscular anatomic characteristics were measured using ultrasonography in 17 studies, computed tomography (CT) in 10 studies and magnetic resonance imaging (MRI) in 8 studies. Fourteen studies were conducted in Europe, 13 in Asia, 4 in Oceania, 3 in Africa and one in the Americas. Four studies classified the patients in sagittal skeletal patterns (Class I, II, III), twenty studies classified

the patients in vertical skeletal patterns (hypodivergent, normodivergent, hyperdivergent) and ten studies classified them both in sagittal and vertical skeletal patterns. Seven studies reported on children, twenty studies on adults, four studies had mixed groups and three studies did not report on the age of their sample. The detailed characteristics of the included studies are presented in detail in Supplementary Tables 6–8.

Risk of bias in individual studies

The risk of bias of the included studies is presented in Supplementary Table 9. In general, the included studies exhibited high risk of bias. Even though the domain of stating aims and objectives was clear, there were several domains that were generally unclear such as the inclusion and exclusion criteria of the samples and the cut-off points of the cephalometric values used to categorize the subjects in the sagittal and vertical skeletal patterns. The domains that were the most problematic were related to the domains of sample size calculation and justification, identification and accounting of confounding factors (age, sex, sagittal classification for studies that grouped per

vertical and vice-versa, functional shifts, asymmetries, linear or curved probe used) in the analyses, incomplete reporting of demographics and results (12 studies with vertical categorization and 5 studies with sagittal categorization reported only correlations between cephalometric and muscular parameters), calibration and experience of the assessors of the muscles. Additionally, most of the studies were unclear with regard to if the assessor of the muscles was blinded to the cephalometric variables.

Results of individual studies and data synthesis

Due to the lack of eligible similar studies from which data could be pulled according to sagittal skeletal categorization, quantitative analysis was feasible only for patients categorized in the vertical dimension.

Descriptive statistics for the cross-sectional area of the masseter muscle (at relaxation and contraction), lateral pterygoid muscle, medial pterygoid muscle, and temporalis muscle are described in Table 1. Descriptive statistics for the volume of the masseter muscle (at relaxation and contraction), lateral pterygoid muscle and medial pterygoid muscle are described in Table 2. Descriptive

Table 1 Descriptive statistics for the muscle CSAs per vertical group

Outcomes	Norm		Hypo		Hyper	
	N, n (min–max)	Pooled mean (95% CI); I ²	N, n (min–max)	Pooled mean (95% CI); I ²	N, n (min–max)	Pooled mean (95% CI); I ²
Masseter CSA (mm ²)	6; 115 [4–35]	3.8 (3.1–4.5); I ² = 0.95	4; 62 [10–22]	4.1 (3.2–5.1); I ² = 0.98	6; 75 [5–25]	3.3 (2.8–3.8); I ² = 0.92
Lateral pterygoid CSA (mm ²)	1; 35 [35–35]	4.2 (4.0–4.4)	0; 0		1; 13 [13–13]	3.7 (3.4–4.0)
Medial pterygoid CSA (mm ²)	3; 79 [11–35]	2.9 (2.6–3.2); I ² = 0.85	2; 30 [11–19]	2.9 (2.6–3.2); I ² = 0.57	3; 46 [8–25]	2.5 (2.2–2.9); I ² = 0.77
Temporalis CSA (mm ²)	1; 35 [35–35]	5.1 (4.9–5.4)	0; 0		1; 13 [13–13]	4.4 (4.1–4.8)
Masseter CSA_bite (mm ²)	1; 10 [10–10]	3.3 (2.7–3.9)	1; 10 [10–10]	4.1 (3.8–4.4)	1; 10 [10–10]	3.1 (2.7–3.5)

N number of studies reporting the outcome in more than 1 patient; n total number of patients in the studies; min–max number of patients in the smallest and the largest study; I² statistic for assessing the between-study heterogeneity (from 0 (no heterogeneity) to 1 (extreme heterogeneity))

Table 2 Descriptive statistics for the muscle volumes per vertical group

Outcomes	Norm		Hypo		Hyper	
	N, n (min–max)	Pooled mean (95% CI); I ²	N, n (min–max)	Pooled mean (95% CI); I ²	N, n (min–max)	Pooled mean (95% CI); I ²
Masseter volume (cm ³)	6; 99 [4–33]	17.2 (14.7–19.7); I ² = 0.96	6; 88 [5–22]	20.8 (15.7–25.9); I ² = 0.98	7; 89 [5–25]	15.7 (12.9–18.4); I ² = 0.95
Lateral pterygoid volume (cm ³)	1; 9 [9–9]	7.0 (6.3–7.8)	1; 5 [5–5]	6.57 (6.1–7.0)	1; 6 [6–6]	5.7 (5.1–6.4)
Medial pterygoid volume (cm ³)	3; 53 [9–33]	8.1 (6.3–10.0); I ² = 0.92	3; 35 [5–19]	8.2 (5.3–11.2); I ² = 0.92	3; 39 [6–25]	7.2 (4.4–10.0); I ² = 0.91
Masseter volume_bite (cm ³)	1; 20 [20–20]	14.9 (14.3–15.5)	1; 20 [20–20]	16.7 (16.2–17.2)	1; 20 [20–20]	12.7 (12.2–13.2)

N number of studies reporting the outcome in more than 1 patient; n total number of patients in the studies; min–max number of patients in the smallest and the largest study; I² statistic for assessing the between-study heterogeneity (from 0 (no heterogeneity) to 1 (extreme heterogeneity))

Table 3 Descriptive statistics for the angle of insertion of the masseter muscle relative to Frankfort horizontal plane per vertical group

Outcomes	Norm		Hypo		Hyper	
	N, n (min–max)	Pooled mean (95% CI); I ²	N, n (min–max)	Pooled mean (95% CI); I ²	N, n (min–max)	Pooled mean (95% CI); I ²
Masseter angle (°)	3; 103 [9–83]	69.3 (64.2–74.4); I ² = 0.95	3; 81 [5–65]	71.4 (69.2–73.5); I ² = 0.66	3; 48 [6–34]	65.2 (59.3–71.1); I ² = 0.86

N: number of studies reporting the outcome in more than 1 patient; n: total number of patients in the studies; min–max: number of patients in the smallest and the largest study; I²: statistic for assessing the between-study heterogeneity (from 0 (no heterogeneity) to 1 (extreme heterogeneity))

statistics for the angle of insertion of the masseter muscle relative to the Frankfort horizontal plane are described in Table 3. Descriptive statistics for the width and thickness of the masseter muscle (at relaxation and contraction) are described in Table 4.

According to the meta-analysis, considerable differences were seen in the masseter muscle cross-sectional area (CSA), volume, thickness in relaxation and thickness in contraction among the three vertical groups. Regarding the comparisons between hypodivergent and normodivergent patients, hypodivergent patients had significantly greater masseter CSA by 0.50 mm² (95% CI 0.05–0.95 mm²); volume by 1.65 cm³ (95% CI 0.45–2.85

cm³); masseter thickness at relaxation by 1.14 mm (95% CI 0.74–1.53 mm); and masseter thickness at contraction by 1.61 mm (95% CI 0.96–2.27 mm). Regarding the comparisons between hyperdivergent and normodivergent patients, hyperdivergent patients had significantly decreased masseter CSA by –0.54 mm² (95% CI –0.95 to –0.12 mm²); volume by –2.64 cm³ (95% CI –3.90 to –1.38 cm³); masseter thickness at relaxation by –1.14 mm (95% CI –1.56 to –0.73 mm); and masseter thickness at contraction by –1.00 mm (95% CI –1.65 to –0.35 mm). Regarding the comparisons between hyperdivergent and hypodivergent patients, hyperdivergent patients had significantly decreased masseter CSA

Table 4 Descriptive statistics for the width and thickness of the masseter muscle per vertical group

Outcomes	Norm		Hypo		Hyper	
	N, n (min–max)	Pooled mean (95% CI); I ²	N, n (min–max)	Pooled mean (95% CI); I ²	N, n (min–max)	Pooled mean (95% CI); I ²
<i>All patients</i>						
Masseter thickness (mm)	9; 202 [4–55]	11.6 (10.4–12.7); I ² = 0.98	9; 152 [10–28]	12.9 (11.6–14.1); I ² = 0.98	9; 165 [5–32]	10.5 (9.1–12.0); I ² = 0.98
Masseter width (mm)	2; 24 [4–20]	41.6 (33.5–49.7); I ² = 0.79	1; 20 [20–20]	46.0 (44.2–47.7)	2; 25 [5–20]	39.3 (36.8–41.7); I ² = 0.3
Masseter thickness _bite (mm)	7; 183 [10–55]	12.9 (11.6–14.2); I ² = 0.97	8; 137 [10–28]	14.7 (13.4–15.9); I ² = 0.97	8; 160 [10–32]	12.2 (10.8–13.7); I ² = 0.98
Masseter width _bite (mm)	1; 20 [20–20]	40.0 (38.2–41.7)	1; 20 [20–20]	42.0 (40.7–43.3)	1; 20 [20–20]	37.0 (35.7–38.3)
<i>Males</i>						
Masseter thickness (mm)	4; 50 [6–26]	13.2 (11.7–14.6); I ² = 0.84	4; 38 [4–17]	14.7 (14.0–15.3); I ² = 0.39	4; 38 [8–11]	12.2 (9.9–14.5); I ² = 0.94
Masseter width (mm)	0; 0		0; 0		0; 0	
Masseter thickness _bite (mm)	4; 50 [6–26]	14.7 (13.7–15.7); I ² = 0.63	4; 38 [4–17]	16.1 (15.6–16.7); I ² = 0	4; 38 [8–11]	13.7 (11.2–16.3); I ² = 0.95
Masseter width _bite (mm)	0; 0		0; 0		0; 0	
<i>Females</i>						
Masseter thickness (mm)	4; 66 [8–29]	12.1 (11.7–12.6); I ² = 0	4; 35 [7–11]	12.8 (11.6–13.9); I ² = 0.72	4; 61 [8–24]	11.5 (10.5–12.5); I ² = 0.71
Masseter width (mm)	0; 0		0; 0		0; 0	
Masseter thickness _bite (mm)	4; 66 [8–29]	13.5 (13.0–13.9); I ² = 0	4; 35 [7–11]	14.5 (14.1–15.0); I ² = 0.4	4; 61 [8–24]	12.9 (12.1–13.7); I ² = 0.62
Masseter width _bite (mm)	0; 0		0; 0		0; 0	

N: number of studies reporting the outcome in more than 1 patient; n: total number of patients in the studies; min–max: number of patients in the smallest and the largest study; I²: statistic for assessing the between-study heterogeneity (from 0 (no heterogeneity) to 1 (extreme heterogeneity))

by -1.04 mm^2 (95% CI -1.49 to -0.59 mm^2); volume by -4.29 cm^3 (95% CI -5.52 to -3.06 cm^3); masseter thickness at relaxation by -2.28 mm (95% CI -2.71 to -1.85 mm); and masseter thickness at contraction by -2.61 mm (95% CI -3.26 to -1.97 mm) (Table 5, Figs. 2, 3, 4, and 5).

A leave-one-out sensitivity analysis was conducted for each measured dimension. When the study by Tekucheva et al. [40] was removed from the analysis of CSA, the between-study variability (Tau^2) decreased from 0.61 to 0.37. However, the heterogeneity was still statistically significant ($p < 0.0001$), and the difference between groups remained significant ($p < 0.0001$). The same occurred when excluding the study by Gregor et al. [41] (which comprises a pool of male patients only); the between-study variability (Tau^2) decreased from 21.32 to 8.43, but the heterogeneity was still significant ($p < 0.0001$), and the difference between groups remained statistically significant ($p < 0.0001$). Regarding muscle thickness, the studies by Lione et al. [42] and Noviello et al. [43] (composed of the same sample of patients) reported lower values than other studies. If these are excluded from the statistical analysis (both for the resting and contraction conditions), Tau^2 decreased from 3.36 to 1.69. However, the heterogeneity and difference between groups remained statistically significant ($p < 0.0001$).

Fourteen studies categorized the included patients according to their sagittal skeletal relationships as Class I, II and III. Due to heterogeneity and incomplete reporting of the outcomes, meta-analysis could

not be performed. Arijji et al. [44] compared the masseter muscle angle of insertion and CSA between Class I and Class III patients and found that the inclination was significantly decreased, and the CSA significantly increased in the Class I group. Another study [45] reported the anatomical characteristics (length and angle) of the masseter muscle in the three different classes of patients (Class I; II; III). Their results revealed that the most acute orientation angle ($67.2 \pm 6.6^\circ$) was found in Class II subjects, while the most obtuse orientation angle ($81.6 \pm 6.8^\circ$) was observed in Class III group; however, no significant differences were found in muscle length among the three groups. Kim et al. [46] reported masseter, medial pterygoid, lateral pterygoid and temporalis muscle thickness separately in males and females in class I and class III patients. There was a significant negative correlation only between masseter muscle thickness in Class III patients and the ANB angle. Kim et al. [47] compared masseter muscle volume/length ratio between Class I and Class III patients and this was significantly greater in Class I patients. Rani and Ravi [32] report the masseter thickness of class I patients and class II patients, making a distinction between patients with maxillary excess and patients with mandibular growth deficiency. The masseter thickness of Class I patients was similar to Class II patients with maxillary excess but class II patients with mandibular deficiency showed thinner masseter thicknesses. Zepa et al. [48] compared the anatomical characteristics (CSA; thickness; volume; length; and width)

Table 5 Pairwise comparisons of masseter muscle cross-sectional area (CSA), volume, thickness at relaxation and at maximal bite per vertical categorization

Outcome	Hypo versus Norm*		Hyper versus Norm**		Hyper versus Hypo***		Heterogeneity**** (p value)	Tau^2	Overall p value*****
	Pooled MD (95%CI)	p value	Pooled MD (95%CI)	p value	Pooled MD (95%CI)	p value			
Masseter CSA (mm^2)	0.50 (0.05–0.95)	0.0289	–0.54 (–0.95 to –0.12)	0.0110	–1.04 (–1.49 to –0.59)	<0.0001	<0.0001	0.61	0.0002
Masseter volume (cm^3)	1.65 (0.45–2.85)	0.0070	–2.64 (–3.90 to –1.38)	<0.0001	–4.29 (–5.52 to –3.06)	<0.0001	<0.0001	21.32	<0.0001
Masseter thick- ness (mm)	1.14 (0.74–1.53)	<0.0001	–1.14 (–1.56 to –0.73)	0.0001	–2.28 (–2.71 to –1.85)	<0.0001	<0.0001	3.38	<0.0001
Masseter thick- ness_bite (mm)	1.61 (0.96–2.27)	<0.0001	–1.00 (–1.65 to –0.35)	0.0026	–2.61 (–3.26 to –1.97)	<0.0001	<0.0001	3.36	<0.0001

Statistically significant results at the 5% level

MD mean difference, CI confidence interval

*Positive mean difference = mean higher in hypo than in norm

**Positive mean difference = mean higher in hyper than in norm

***Positive mean difference = mean higher in hyper than in hypo

****Residual heterogeneity

*****For testing the equality of the mean in the three groups

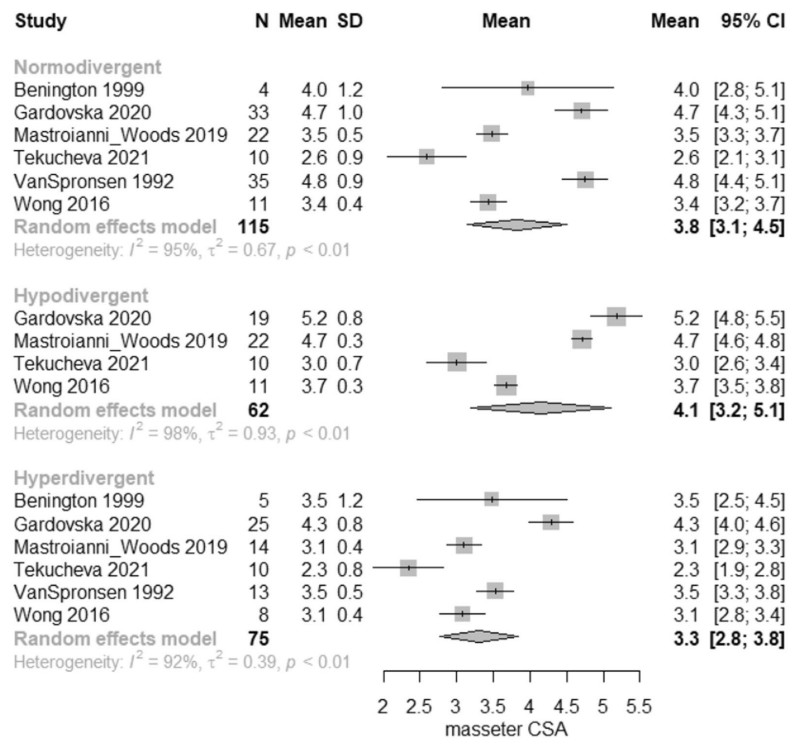


Fig. 2 Forest plot of masseter cross-sectional area (CSA) in normodivergent, hypodivergent and hyperdivergent patients

of the masseter and medial pterygoid muscles between Class II and Class III patients. In Class III patients, there was a tendency for all masseter variables to be higher; however, they did not reach statistical significance. On the contrary, the volume and the thickness of the medial pterygoid muscles were significantly greater in the Class III patients compared to Class II patients.

Five studies reported only correlations between muscular anatomic characteristics and sagittal skeletal classification with mainly insignificant findings (Supplementary Table 7). The certainty of the evidence according to the GRADE rating was judged as being very low, and the reasons for downgrading were study design (observational, cross-sectional), individual study limitations due to high risk of bias, inconsistency of the results due to great heterogeneity, and imprecision due to small sample sizes and wide CIs (Supplementary Tables 10–12).

Discussion

General interpretation of the results in the context of other evidence

The results of the present systematic review and meta-analysis show significant differences in the parameters related to masseter muscle volume, CSA, width and thickness across different groups of patients categorized by their facial vertical divergence. In general, it was shown that hyperdivergent patients had smaller muscles for all analysed outcomes compared to normodivergent and hypodivergent patients while normodivergent patients had smaller muscles compared to hypodivergent patients. These results underpin a possible association between masticatory muscular characteristics and vertical craniofacial morphology; however, the magnitude of the differences was small and the certainty of the evidence very low.

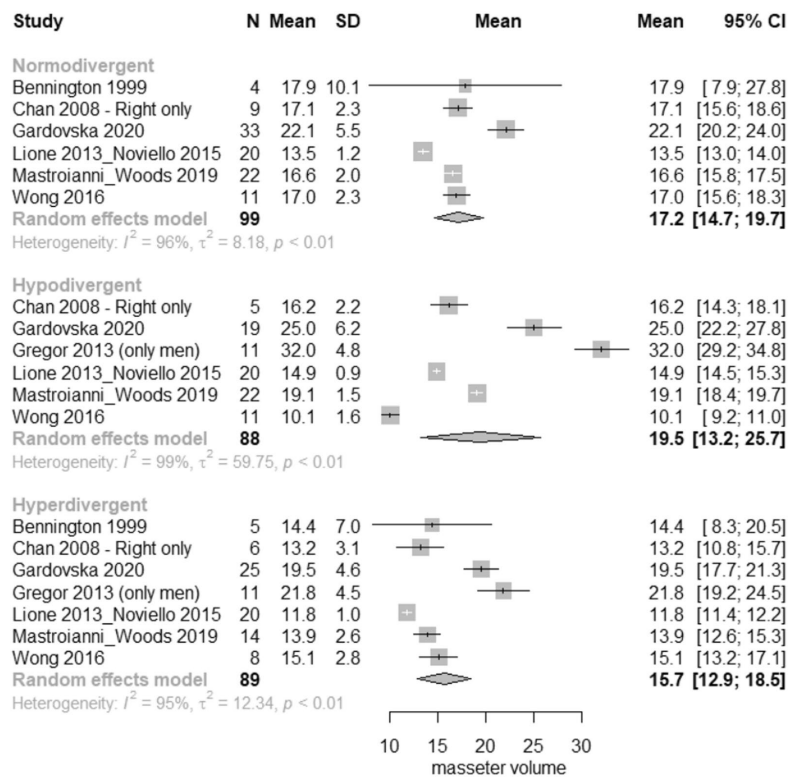


Fig. 3 Forest plot of masseter volume in normodivergent, hypodivergent and hyperdivergent patients

Alternative approaches for assessing the functional capacity of masticatory muscles have been explored concerning vertical craniofacial morphology, yielding largely consistent findings. Studies have demonstrated a negative association between vertical facial dimensions and maximal bite force as well as electromyographic activity of masticatory muscles [49, 50]. Similarly, findings from computer tomography (CT) investigations align with these results, indicating a negative correlation between the mandibular plane angle and parameters such as masseter muscle thickness and length [51].

It has been previously reported that muscular anatomic characteristics are also reflected in the forces they are capable to exert with thicker muscles being able to deliver greater mechanical stresses on the underlying skeletal bone structures [52, 53]. Similarly, the masticatory muscles and their constraints exerted on facial bone structures considerably influence the face in general

and mandibular shape [54, 55]. Additionally, it has been found that women with thinner muscles have longer faces while subjects with muscular dystrophies and subsequently weaker muscles attain a hyperdivergent growth pattern, which in turn indicates a relationship between aberrations in muscular characteristics and deviations in craniofacial morphology [20, 56].

The distribution of muscle fibres and their molecular structure also differs between vertical and horizontal growers. Studies have shown that the more hyperdivergent the patients, the more type I muscle fibres (slow fibres) are found, while in hypodivergent patients type II fibres (fast fibres) are more numerous [57, 58]. The correlation between sagittal jaw relationships and mean fibre area was found to be less evident; however, within Class III subjects, those with a deep bite exhibited a notable rise in type I and I/II hybrid fibres while polymorphism in the MYO1H gene was linked to an

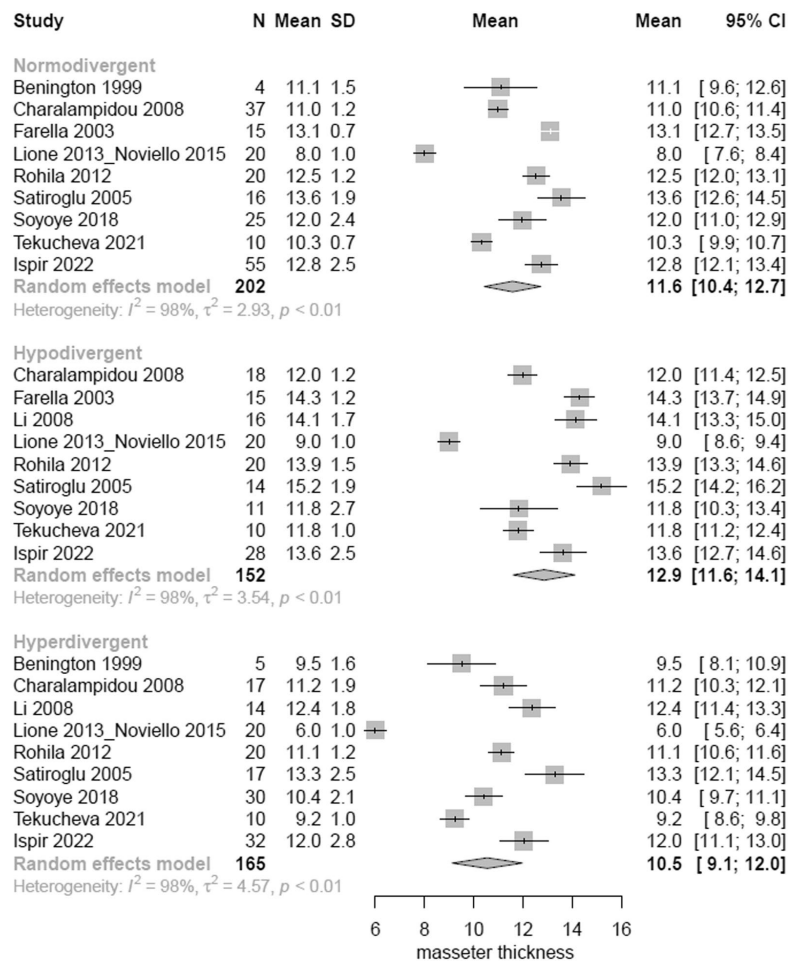


Fig. 4 Forest plot of masseter thickness at relaxation in normodivergent, hypodivergent and hyperdivergent patients

elevated susceptibility to mandibular prognathism and horizontal maxillomandibular discrepancies irrespective to the ethnic background [57, 59]. Associations between fibre-type distribution and biochemical composition of the masticatory muscles and how these can relate to skeletal craniofacial patterns however warrant further investigation.

Limitations of the evidence included in the review and the review process

Even though the present protocol was pre-registered, and an exhaustive literature search was performed, this systematic review also comes with some limitations relevant mainly to its results and the high risk of bias of the included studies. Issues related to bias are pertinent to study size (sample size), inconsistencies in the cut-off values of the cephalometric parameters used to categorize the patients into sagittal and/or vertical groups, unclear

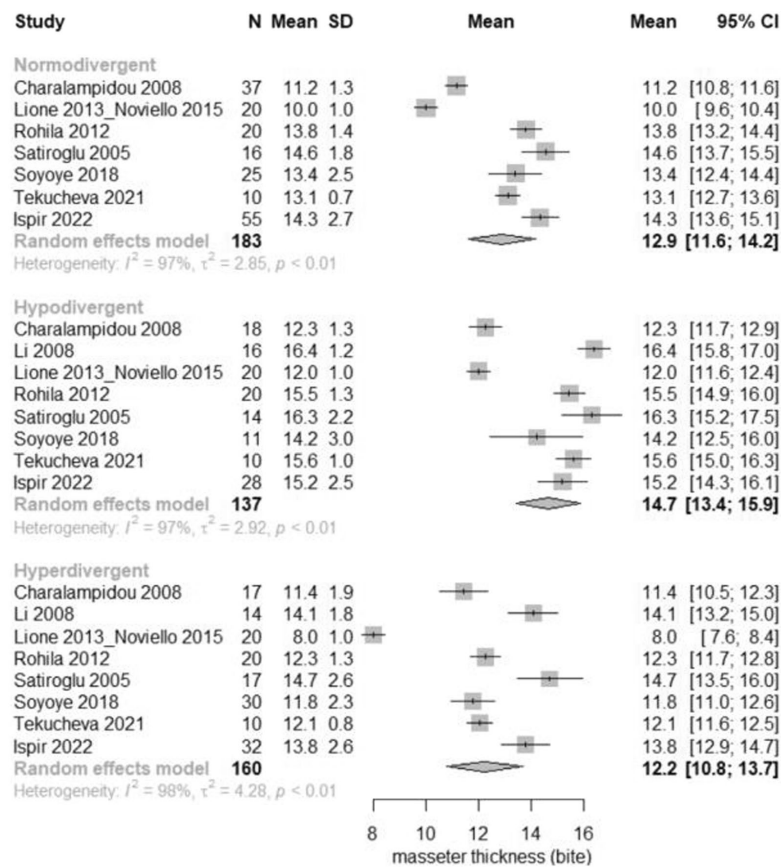


Fig. 5 Forest plot of masseter thickness at bite in normodivergent, hypodivergent and hyperdivergent patients

information on whether the assessor of the muscles was trained and blinded to the cephalometric variables and incomplete reporting of the samples and results.

Various confounding factors were also not taken into consideration and accounted in the analyses in the included studies such as patients' age and sex. Several studies included only males in their sample, naturally presenting larger muscle dimensions than those of women [24, 26, 41, 60]. Given that outcomes related to muscular characteristics differ between vertical groups, such subcategorization should be accounted for when grouping the patients per sagittal group. Additionally, age also varied greatly between the samples of different studies. Knowing that muscular bite force differences

are not evident between hyperdivergent and normodivergent patients during childhood but only after adulthood, carrying out analyses without accounting for age can bias the results, however, such sub-group analyses were not possible due to incomplete reporting of study samples and results [13].

It could be advocated that differences in image acquisition could impart the accuracy of the measurements; however, the studies that contributed to the meta-analyses assessed muscular characteristics primarily by using ultrasonography. Moreover, studies show no significant differences between different methods such as Cone-Beam CT, MRI scans and ultrasonography for

measuring and analysing muscle characteristics [61, 62].

When considering masticatory muscle size assessments, those made under relaxed conditions are known to be less reproducible due to the fact that the relaxed muscles are more susceptible to the pressure with which the transducer is positioned against the cheek, and thus is very technique sensitive [31, 63–65]. Measurements made under contracted conditions (with the patient biting) are thus preferred when assessing masticatory muscle thickness characteristics.

Finally, the overall high risk of bias of the included studies precludes us from drawing robust conclusions based on the current available evidence.

Implications of the results for practice, policy, and future research

These results highlight the importance of masticatory muscles in shaping the skeletal structures of the face. Even though in the past orthodontic treatment was basically focused on dental relationships, the current trends are more towards face-oriented orthodontic treatments [66]. Additionally, muscular anatomic characteristics have been reported to influence response to orthodontic treatment [67]. More specifically, the initial thickness of the masseter muscles in patients treated with functional appliances has been shown to influence treatment outcome. There was greater posterior displacement of the cephalometric point A in patients with thinner masseter muscles but also greater mandibular incisor proclination [68]. After undergoing functional appliance therapy, children exhibiting more pronounced dentoalveolar changes (thinner masseter muscles) may also demonstrate an increased likelihood of sagittal relapse post-treatment [69].

Although the current scientific evidence on the role of masticatory muscles on craniofacial patterns is weak, it could highlight an additional approach to patient care by incorporating factors related to muscular characteristics in baseline diagnosis. This approach considers the physiological aspects of a malocclusion, establishing the biological limits within which a practitioner can work. This applies to all treatment methods, whether conventional, combined with orthognathic surgery, using various appliances and biomechanics, or involving retention strategies for maintaining long-term stability.

With the present systematic review and meta-analysis, an attempt was made to gather the available evidence regarding the differences in anatomic masticatory muscle characteristics and craniofacial patterns as categorized in the sagittal or vertical dimensions using cephalometry. Given the uncertainty of the evidence though regarding the differences found in the present review, it is

recommended that further high-quality prospective studies are conducted to expand the available evidence in this field.

Conclusions

Based on the studies included in our systematic review and meta-analysis, masseter muscle volume, cross-sectional area, width and thickness (under both relaxation and contraction) were significantly decreased in hyperdivergent patients compared to normodivergent and hypodivergent while the same parameters were significantly increased in hypodivergent patients compared to normodivergent patients. These results should be interpreted with caution because the scientific evidence from primary studies is weak with a high risk of bias.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40510-024-00534-2>.

Additional file1 (DOCX 1154 KB)

Additional file2 (DOCX 464 KB)

Acknowledgements

The authors would like to thank Mr Christophe Combescur, Service d'épidémiologie Clinique, Centre de recherche Clinique, University of Geneva for the assistance with the meta-analyses.

Funding

Open access funding provided by University of Geneva.

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Received: 9 April 2024 Accepted: 10 July 2024

Published online: 09 September 2024

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Discussion

Interprétation générale des résultats

Les résultats de cette revue systématique et méta-analyse montrent des différences significatives dans les caractéristiques musculaires liées au volume, à la surface transversale, à la largeur et à l'épaisseur du muscle masséter dans différents groupes de patients classés en fonction de leur divergence faciale.

Une possible association entre phénotype musculaire et morphologie cranio-faciale peut donc être évoquée car, de manière générale, les patients ayant un visage allongé présentent des muscles masséters plus fins alors que les patients ayant des visages courts présentent quant à eux une épaisseur musculaire plus importante.

Comme évoqué précédemment dans l'introduction, une contrainte mécanique exercée sur une structure osseuse va influencer son remodelage et sa morphologie. De ce fait, des muscles masticatoires puissants exerçant une force majeure par rapport à des muscles moins épais [54,55], induisent considérablement plus de remodelage. Ce remodelage se traduit alors par les caractéristiques morphologiques typiques des patients hypodivergents tels qu'un angle gonial et l'angle du plan mandibulaire aigus [56,57]. L'association entre typologie musculaire et déviation de la morphologie cranio-faciale évoquée dans cette discussion est renforcée par d'autres résultats cliniques qui montrent que les femmes, ayant en moyenne des muscles masséters plus fins que les hommes, ont généralement un visage plus allongé [24]. De même, les patients atteints de dystrophie musculaire présentent eux aussi un modèle de croissance cranio-faciale vertical [58].

On trouve également dans la littérature quelques études qui se sont intéressées à la relation entre morphologie cranio-faciale et caractéristiques anatomiques microscopiques des muscles masticatoires, telles que la distribution des différentes fibres musculaires et la composition

moléculaire. En effet, les patients hyperdivergents auraient majoritairement des fibres musculaires de type I (lentes), alors que les patients avec des visages moins allongés seraient dotés de plus de fibres de type II (rapides) [59,60]. La proportion de distribution de ces fibres est moins évidente pour la classification sagittale des patients, mais le polymorphisme du gène MYO1H est lié à une susceptibilité élevée au prognathisme mandibulaire [61]. Malgré ces quelques publications, la relation entre morphologie cranio-faciale et caractéristiques microscopiques des muscles masticatoires reste peu étudiée, ce qui pourrait mériter de plus amples recherches à l'avenir.

Limites des données incluses dans la revue

Différents facteurs de biais n'ont pas été pris en considération lors de la réalisation des études incluses, tels que l'âge et le sexe des patients. En effet l'âge varie considérablement entre les différentes études : certaines études incluent uniquement des enfants, d'autres uniquement des adultes, quelques-unes incluent à la fois des adultes et des enfants et parfois l'âge des patients examinés n'est pas rapporté. Sachant que peu de différences existent lors de la contraction maximale des muscles masticatoires chez les enfants hyperdivergents et normodivergents, mais que ces différences se manifestent uniquement à l'âge adulte [33], effectuer des analyses sans tenir compte de l'âge peut fausser les résultats.

Plusieurs études n'ont inclus que des hommes dans leur échantillon, ce qui biaise inévitablement aussi les résultats car ils présentent naturellement des dimensions musculaires plus importantes que celles des femmes.

De plus, pour diminuer les biais des études incluses, une sous-catégorisation aurait dû être prise en compte lors du regroupement des patients par groupe sagittal, étant donné que les résultats liés aux caractéristiques musculaires diffèrent entre les groupes verticaux.

Quant à l'utilisation de différentes typologies d'acquisition des images (CBCT, IRM, CT, US), la littérature ne montre pas de différences significatives entre elles pour mesurer et analyser les caractéristiques musculaires [62,63]. La majorité des études ont cependant utilisé l'ultrasonographie pour des raisons pratiques et économiques.

En ce qui concerne l'évaluation de la taille des muscles masticatoires, l'enregistrement est préférable lorsque le muscle est contracté. En effet un muscle contracté nous donnera des mesures plus reproductibles que les muscles détendus car une pression est exercée par la sonde sur la joue lors de l'enregistrement [13,64–66]. Les études incluses dans notre revue systématique rapportent à la fois de valeurs enregistrées au repos et des valeurs lors de la contraction maximale pour l'enregistrement de la section transversale, épaisseur et volume du masséter.

Enfin, le risque global élevé de biais des études incluses nous empêche de tirer des conclusions solides sur la base des données actuellement disponibles.

Implications des résultats pour la pratique et la recherche future

Ces résultats soulignent l'importance des muscles masticatoires dans la formation des structures squelettiques du visage. Même si, par le passé, le traitement orthodontique était essentiellement axé sur les relations dentaires, la tendance actuelle est plutôt aux traitements orthodontiques axés sur le visage [67]. En outre, il a été rapporté que les caractéristiques anatomiques musculaires influencent la réponse au traitement orthodontique [68]. En effet, il a été démontré que l'épaisseur initiale des muscles masséters chez les patients traités avec des appareils fonctionnels influence le résultat du traitement. Le déplacement postérieur du point céphalométrique A est plus important chez les patients dont les muscles masséters est plus fins, mais la proclinaison des incisives mandibulaires est également plus importante [69]. Après avoir suivi un traitement par appareil fonctionnel, les enfants présentant des

changements dentoalvéolaires plus prononcés (muscles masséters plus fins) peuvent également présenter une probabilité accrue de récurrence sagittale après le traitement [70].

Bien que les preuves scientifiques actuelles concernant le rôle des muscles masticatoires sur les schémas cranio-faciaux soient faibles, elles pourraient mettre en évidence une approche supplémentaire des soins aux patients en incorporant des facteurs liés aux caractéristiques musculaires dans le diagnostic de base.

Cette revue systématique et méta-analyse a tenté de rassembler toutes les preuves scientifiques disponibles concernant les différences entre les caractéristiques anatomiques macroscopiques des muscles masticatoires et les caractéristiques de la morphologie faciale sur le plan sagittal et vertical à l'aide de la céphalométrie. Étant donné l'incertitude des preuves scientifiques dans cette revue systématique et méta-analyse, il est recommandé de mener de nouvelles études prospectives de meilleure qualité.

Conclusions

Malgré la faible preuve scientifique associée aux résultats de cette méta-analyse, ils révèlent une différence significative dans l'épaisseur des muscles masséters entre les patients hyperdivergents, normodivergents et hypodivergents. En effet, les muscles masséters des patients hyperdivergents sont significativement moins épais que ceux des patients normodivergents et hypodivergents, tant au repos que lors de leur contraction maximale. De plus, le volume et la surface transversale du masséter chez les sujets hypodivergents sont significativement plus importants que chez les sujets normodivergents et hyperdivergents.

Remerciements

Je tiens à remercier la Docteure Alexandra Papadopoulou et le Professeur Gregory Antonarakis pour leur soutien, leur disponibilité, leur encouragement et leur très précieuse aide tout au long de cette thèse.

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